# CHARLES ROTICH

**Computational Physicist | Materials Science Researcher**

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## PROFESSIONAL PROFILE

Computational physicist and materials-science researcher with training in first-principles modelling, high-performance computing, cloud computing and materials testing. Pursuing an MSc in Physics (Materials Science) at The Technical University of Kenya with previous experience as a Graduate Research Assistant. Research interests include electronic structure, band structures and DOS/PDOS, band-gap formation in perovskite oxides, photocatalytic water splitting, and temperature and vacancy effects on mechanical properties. Experienced with Linux-based scientific computing and atomistic simulation workflows.

## EDUCATION

**MSc in Physics (Materials Science)** | *The Technical University of Kenya*
**2024 - Present** | Kenya

- Current postgraduate research focuses on computational materials science and first-principles modelling.

**Bachelor of Technology in Technical and Applied Physics** | *The Technical University of Kenya*
**2017 - 2022** | Kenya

- First Class Honours.
- Specialization included materials science and computational physics.

## RESEARCH EXPERIENCE

**Graduate Research Assistant** | *The Technical University of Kenya*
**Jul 2025 - Aug 2026** | Nairobi, Kenya

- Conducted computational materials research using first-principles and atomistic modelling approaches.
- Supported research activities in computational physics and materials science.

**Researcher (Internship)** | *Kenya Education Network Trust (KENET)*
**Apr 2023 - Jul 2025** | Nairobi, Kenya / Remote

- Tested and developed use cases for the KENET Research Cloud platform.
- Applied computational physics and high-performance computing experience to research-cloud workflows.
- Worked with multidisciplinary teams and supported researchers using scalable computational resources.

**Research Physicist (Internship)** | *TUK Material Modelling Team*
**Apr 2023 - Jul 2025** | Nairobi, Kenya

- Supported computational materials research using SIESTA and Quantum ESPRESSO.
- Guided students in the use of electronic-structure simulation tools and research workflows.
- Investigated mechanical, electronic, and optical properties of materials using computational methods.
- Presented research work in national and international scientific settings.

**Physicist (Attachment/Internship)** | *Materials Testing and Research Division*
**Jan 2022 - May 2022** | Nairobi, Kenya

- Performed non-destructive testing using radiography, ultrasonic testing, and visual inspection.
- Conducted destructive materials tests including hardness, impact, and tensile testing.
- Contributed to development of an instrument for calibration of rebar detectors.

## MSC RESEARCH PROJECT

### First-Principles Investigation of Electronic, Optical, Redox Properties of Spinel Oxides $CoMn_2O_4$ and $CuCo_2O_4$ for Photocatalytic Water Splitting

- Study of bulk and surface properties of spinel oxides using density functional theory.
- Research scope includes structural stability, electronic structure, optical response, surface stability, adsorption of O*, OH*, OOH* and $H_2O$, and oxygen-evolution reaction energetics.
- Uses computational hydrogen electrode concepts to assess reaction free-energy pathways and catalytic overpotential.
- Computational tools include VASP and Quantum ESPRESSO, with Linux/HPC workflows for large-scale calculations.

## BACHELOR RESEARCH PROJECT

**Effects of Temperature and Intrinsic Structural Defects on Mechanical Properties of Niobium Nitride**

Bachelor of Technology final-year project | The Technical University of Kenya

*Report dated September 2021*

- Studied temperature and nitrogen-vacancy effects on rock-salt NbN using Quantum ESPRESSO and molecular dynamics, with convergence tests and structural relaxation.
- Used AiiDA and ASE to handle workflows and structures, and Birch-Murnaghan fits to compare bulk moduli of pristine, defective and thermally relaxed NbN.

## INTERNATIONAL RESEARCH & ACADEMIC MOBILITY

**Erasmus+ KA171 Research Mobility** | *University of Cantabria*
**2026 - 2027** | Santander, Spain

- Five-month international academic and research exchange undertaken through the Erasmus+ partnership with The Technical University of Kenya.
- Mobility supports advanced research experience and international collaboration in computational/materials physics.

## ADVANCED SCHOOLS & SCIENTIFIC TRAINING

**APhRICA Advanced Physics Training** | *University of Rwanda / ICTP-EAIFR*
**Multi-year participation (2 years)** | Kigali, Rwanda

- Participated in advanced physics training activities aimed at strengthening postgraduate-level theoretical and computational physics skills.
- ICTP Physics Without Frontiers workshop, Nairobi - training in density functional theory fundamentals (2025).
- DFT School, Catholic University of Eastern Africa - computational materials/DFT training (2024).
- Data Science Africa, Dedan Kimathi University of Technology, Nyeri - scientific/data-science training (2024).
- ALX AI Career Essentials - applied artificial-intelligence skills and professional development.

## TECHNICAL & COMPUTATIONAL SKILLS

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|---|---|
| **Electronic structure / DFT** | VASP, Quantum ESPRESSO, SIESTA, ABINIT; DFT and DFT+U workflows |
| **Materials modelling** | Bulk and surfaces; bands, DOS/PDOS and optical properties; molecular dynamics, bulk modulus and equation-of-state fitting; adsorption and reaction energetics |
| **HPC & scientific computing** | Linux, Bash/terminal, SSH, MPI, Slurm, high-performance computing, cloud computing |
| **Programming & data** | Python, SQL, C++; scientific data analysis and visualization |
| **Scientific software ecosystem** | AiiDA, VASPKIT, Wannier90 and related computational-materials tools |
| **Materials testing** | Non-destructive testing, radiography, ultrasonic testing, visual inspection, hardness, impact and tensile testing |
| **Productivity** | Microsoft Excel, Word and PowerPoint |

## RESEARCH INTERESTS

Electronic structure of silicon in the diamond structure and benzene: band structures, DOS/PDOS and molecular gaps; band-gap formation in perovskite oxides and the roles of structural and magnetic symmetry; spinel oxides, optical properties, surface chemistry and photocatalytic water splitting; temperature and vacancy effects on the mechanical properties of NbN; scientific computing and reproducible simulations.

## SELECTED PROFESSIONAL STRENGTHS

Scientific research and experimentation; computational workflow development; research computing support; student guidance; technical presentation; interdisciplinary collaboration; scientific communication.

## REFERENCES

*Available on request.*